KARTHIK A/L SUBRAMANIAM

SENIOR MACHINE LEARNING ENGINEER

CONTACT

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SUMMARY

An innovative and results-driven Senior Machine Learning Engineer with a Master's degree in Data Science and over 4 years of experience designing scalable artificial intelligence solutions. Proven track record in developing predictive models, natural language processing (NLP) pipelines, and computer vision applications that drive business value. Adept at bridging the gap between complex data science concepts and actionable business strategies, with a strong focus on deploying models into production environments.

WORK & TECHNICAL PROJECTS

Lead ML Engineer: Predictive Maintenance System for Manufacturing — Inari Amertron Berhad **2023 – Present**

- Designed and deployed an end-to-end IoT predictive maintenance pipeline using Python, AWS SageMaker, and Apache Kafka.
- Developed anomaly detection models using Random Forest and XGBoost to analyze real-time sensor data from factory machinery.
- Reduced unplanned equipment downtime by 28% and saved an estimated RM 1.2M annually in maintenance overheads.

Data Scientist: Customer Sentiment & Churn Prediction Engine — Axiata Digital Labs Malaysia **2021 – 2023**

- Built an NLP pipeline leveraging BERT embeddings to analyze thousands of unstructured customer reviews and support tickets.
- Created a churn prediction dashboard using Tableau, integrating deep learning models to identify at-risk client segments.
- Achieved a 15% increase in customer retention rates within the first two quarters of live deployment.

Computer Vision Lead: Automated Quality Assurance — ViTrox Technologies Sdn. Bhd. **2020 – 2021**

- Containerized the inference application using Docker and orchestrated deployment via Kubernetes to handle high-throughput video streams with sub-second latency.
- Trained and optimized object detection models (YOLOv8) to identify micro-defects on automated optical inspection assembly lines.

SKILLS

- **Programming & Databases:** Python, R, SQL, NoSQL (MongoDB)
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-Learn, Keras, HuggingFace
- **Cloud & MLOps:** AWS (SageMaker, EC2, S3), Docker, Kubernetes, MLflow, CI/CD pipelines
- **Soft Skills:** Agile Project Leadership, Cross-functional Communication, Strategic Problem-Solving

EDUCATION

Monash University Malaysia **2022 – 2024**
Master of Data Science

- CGPA: 3.85 / 4.00
- Thesis: "Optimizing Deep Learning Architectures for Resource-Constrained Edge Devices."

Multimedia University (MMU) Cyberjaya **2018 – 2021**
Bachelor of Computer Science (Hons) in Artificial Intelligence

- CGPA: 3.92 / 4.00
- Award: Dean's List for 6 consecutive semesters.

ADDITIONAL INFORMATION

- **Languages:** English (Fluent), Tamil (Native), Malay (Proficient)